

Hands on 4 - Merge Conflict Resolution

Running the following commands in order:

```
mkdir GitConflictDemo
cd GitConflictDemo
git init
```

```
# Create initial file
echo "Initial Content" > readme.txt
git add readme.txt
git commit -m "Initial commit"
```

```
# Verify repository is clean
git status
```

```
# Create a new branch
git branch GitWork
git checkout GitWork
```

```
# Create hello.xml in GitWork
echo "<message>Hello from GitWork</message>" > hello.xml
git add hello.xml
git commit -m "Added hello.xml in GitWork"
```

```
# Switch back to master
git checkout master
```

```
# Create hello.xml with different content
echo "<message>Hello from Master</message>" > hello.xml
git add hello.xml
git commit -m "Added hello.xml in master"
```

```
# View commit history
git log --oneline --graph --decorate --all
```

```
# Compare branches
git diff master GitWork
```

```
# Merge GitWork into master
```

```
git merge GitWork
```

```
# Resolve the merge conflict manually by editing hello.xml
```

```
# Example final content:
```

```
# <message>Hello from Master</message>
```

```
# <message>Hello from GitWork</message>
```

```
# After editing:
```

```
git add hello.xml
```

```
git commit -m "Resolved merge conflict"
```

```
# Create .gitignore and ignore backup files
```

```
echo "*.orig" > .gitignore
```

```
git add .gitignore
```

```
git commit -m "Added .gitignore"
```

```
# Check repository status
```

```
git status
```

```
# List branches
```

```
git branch
```

```
# Delete merged branch
```

```
git branch -d GitWork
```

```
# View final log
```

```
git log --oneline --graph --decorate
```

Output:

```
GitConflictDemo — -zsh — 80x24

[adityahargunani@Adityas-MacBook GitConflictDemo % echo "*.orig" >> .gitignore ]
[adityahargunani@Adityas-MacBook GitConflictDemo % cat .gitignore ]
*.orig
[adityahargunani@Adityas-MacBook GitConflictDemo % git add .gitignore ]
[adityahargunani@Adityas-MacBook GitConflictDemo % git commit -m "Added .gitignor]
e"
[master d60cf96] Added .gitignore
1 file changed, 1 insertion(+)
create mode 100644 .gitignore
[adityahargunani@Adityas-MacBook GitConflictDemo % git branch ]
GitWork
* master
[adityahargunani@Adityas-MacBook GitConflictDemo % git branch -d GitWork ]
Deleted branch GitWork (was 01ad547).
[adityahargunani@Adityas-MacBook GitConflictDemo % git log --oneline --graph --de]
corate
* d60cf96 (HEAD -> master) Added .gitignore
* 7afccf3 Resolved merge conflict
|\
| * 01ad547 Added hello.xml in GitWork
* | c01bf06 Added hello.xml in master
|/
* 0641a31 Initial commit
adityahargunani@Adityas-MacBook GitConflictDemo %
```